1 - 20. Find the following limits, if they exist. Where appropriate, show algebraic work in correct math form to justify your results.

1. $\lim\limits_{x \to 0^+} \sqrt{x}$

2. $\lim\limits_{x \to 0^-} \sqrt{x}$

3. $\lim\limits_{x \to 0} \sqrt{x}$

4. $\lim\limits_{x \to 9^+} \sqrt{x-9}$

5. $\lim\limits_{x \to 5^-} \sqrt{5-x}$

6. $\lim\limits_{x \to 4^+} \sqrt{4-x}$

7. $\lim\limits_{x \to 5^+} \sqrt{\dfrac{8-x}{x-3}}$

8. $\lim\limits_{x \to 5^-} \sqrt{\dfrac{8-x}{x-3}}$

9. $\lim\limits_{x \to 5} \sqrt{\dfrac{8-x}{x-3}}$

10. $\lim\limits_{x \to 8^+} \sqrt{\dfrac{8-x}{x-3}}$

10. $\lim\limits_{x \to 8^-} \sqrt{\dfrac{8-x}{x-3}}$

11. $\lim\limits_{x \to 8} \sqrt{\dfrac{8-x}{x-3}}$

1 - 20. (continued) Find the following limits, if they exist. Where appropriate, show algebraic work in correct math form to justify your results.

13. $\lim_{x \to 3^+} \sqrt{\dfrac{8-x}{x-3}}$

14. $\lim_{x \to 3^-} \sqrt{\dfrac{8-x}{x-3}}$

15. $\lim_{x \to 10^-} \sqrt{\dfrac{8-x}{x-3}}$

16. $\lim_{x \to 3^+} \dfrac{\sqrt{8-x}}{x-3}$

17. $\lim_{x \to 3^-} \dfrac{\sqrt{8-x}}{x-3}$

18. $\lim_{x \to 3} \dfrac{\sqrt{8-x}}{x-3}$

19. $\lim_{x \to +\infty} \sqrt{\dfrac{8}{x-3}}$

20. $\lim_{x \to -\infty} \sqrt{\dfrac{8}{x-3}}$

1 - 10. Find the following limits, if they exist. Where appropriate, show algebraic work in correct math form to justify your results.

1. $\lim_{x \to 2^-} \dfrac{\sqrt{4-x^2}}{\sqrt{2-x}}$

2. $\lim_{x \to 0} \dfrac{x}{\sqrt{x+25}-5}$

3. $\lim_{x \to 1} \dfrac{x-1}{\sqrt{x^2+3}-2}$